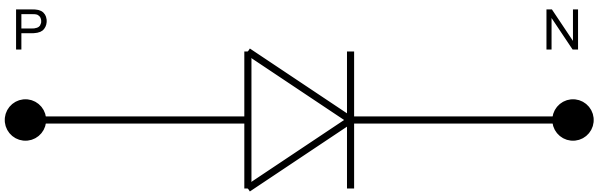
Chapter 3 – The PN Junction

3.1 PN-Junction Electrostatics
3.2 I-V Characteristic
3.3 Dynamic Behavior
3.4 Diode Circuit Models
3.5 Diode Applications and Circuits

Literature:
Pierret, Chapter 5-7
Jaeger Blalock, Chapter 3

The PN Junction

Circuit Symbol

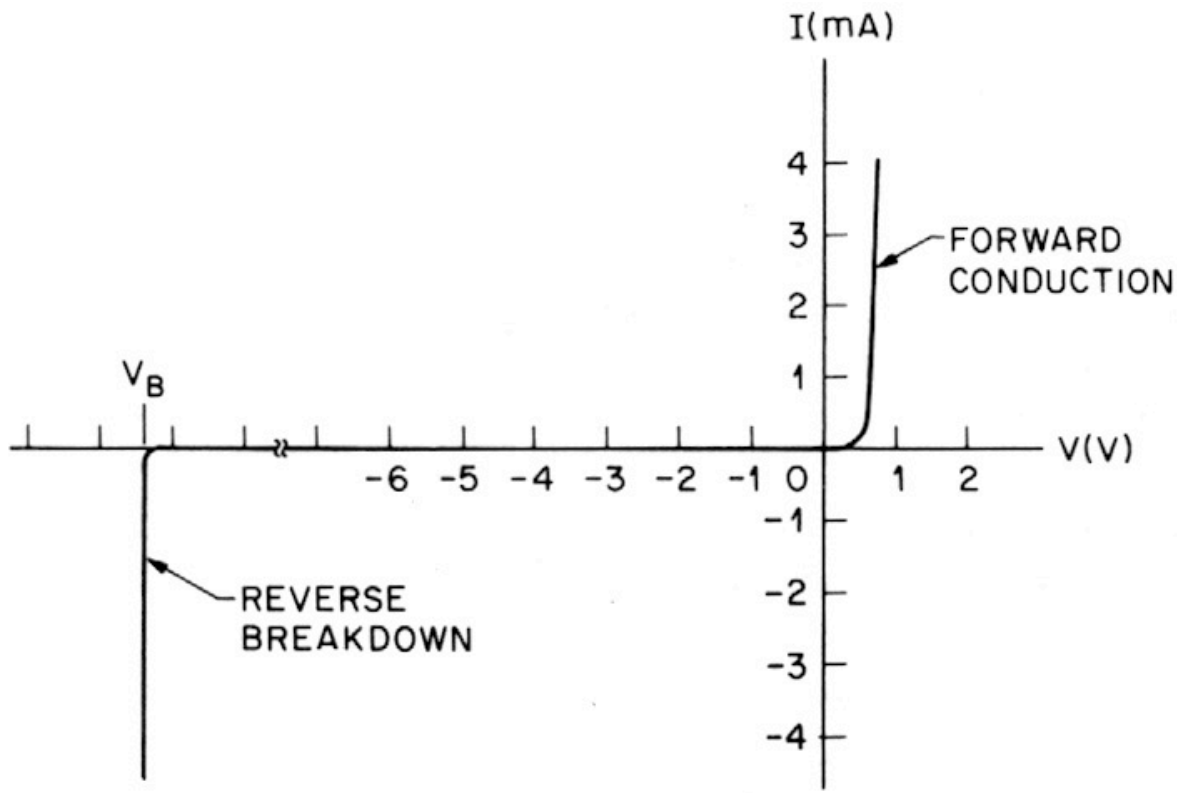


+ Forward Bias –

– Reverse Bias +

$$I = I_s\left[e^{qV_A/kT} - 1\right]$$

I-V Characteristic



3.1 PN Junction: Electrostatics

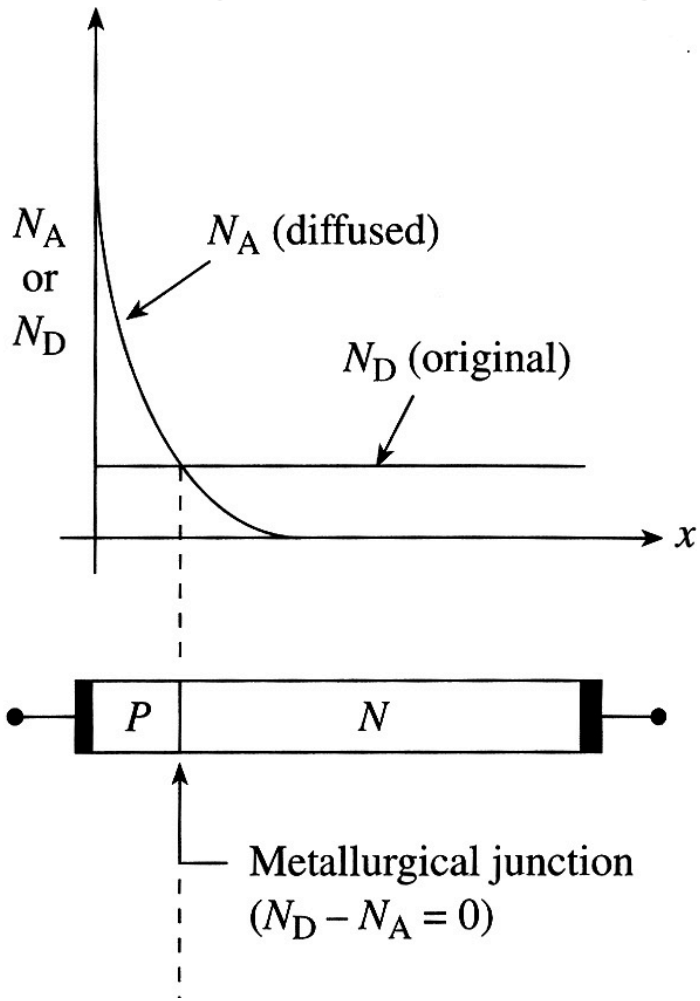
3.1.1 PN Junction Basics
- Junction Approximations
- Band Structure
- Built-In Potential

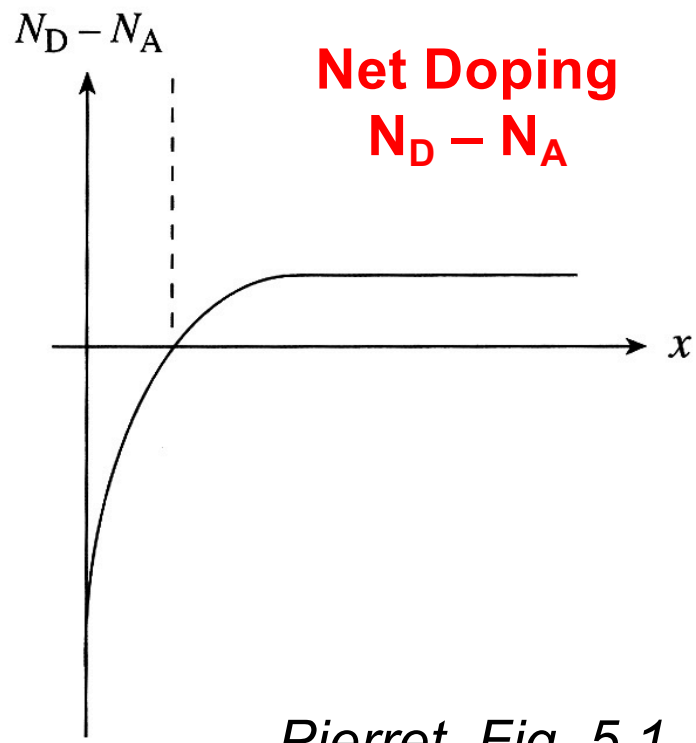
3.1.2 Step PN Junction
- Equilibrium $V_A = 0$
- External Bias $V_A \neq 0$

3.1.3 Linearly Graded PN Junction
- Equilibrium $V_A = 0$
- External Bias $V_A \neq 0$

Pierret, Chapter 5, page 195-226

3.1.1 PN Junction Basics





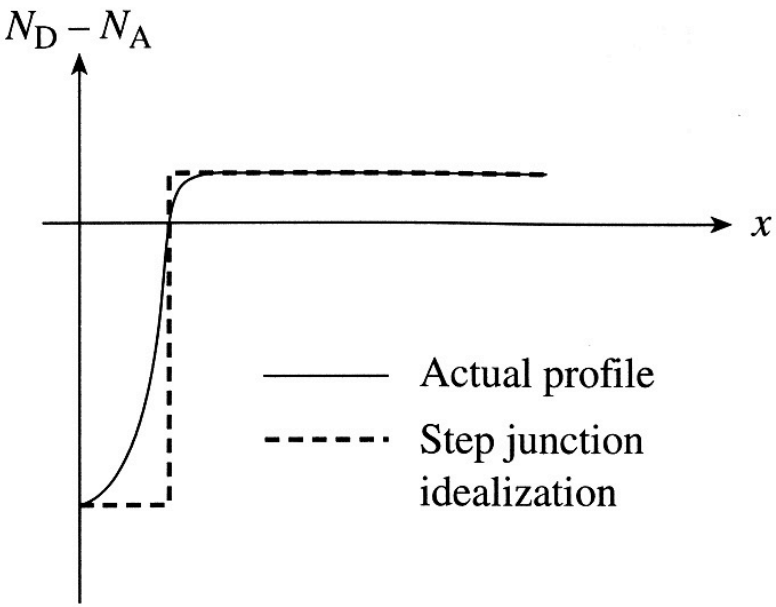
Net Doping
$N_D - N_A$

Pierret, Fig. 5.1

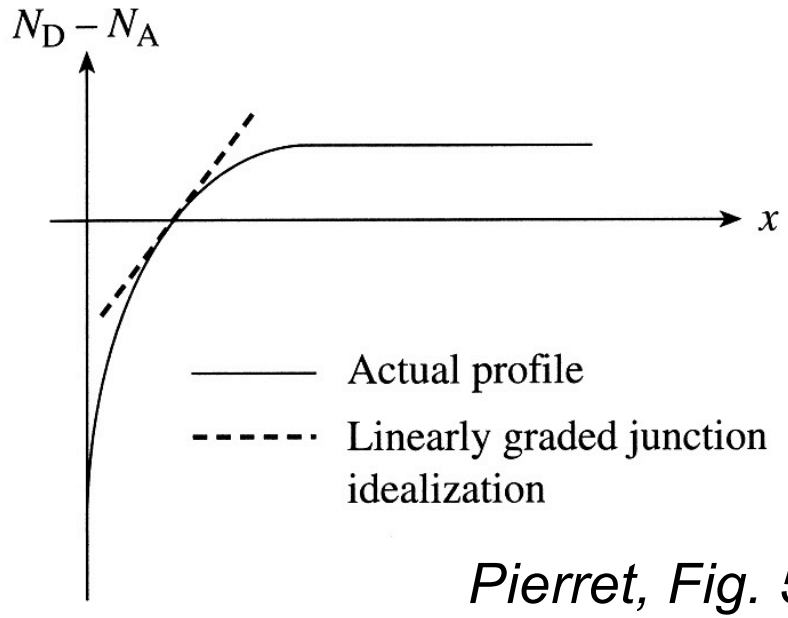
- Example: Diffusion of acceptor atoms (e.g. boron) in n-substrate
- Metallurgical junction at $N_A = N_D$ or $N_D - N_A = 0$

Doping Profile Approximations for PN Junctions

Step Junction



Linearly Graded Junction



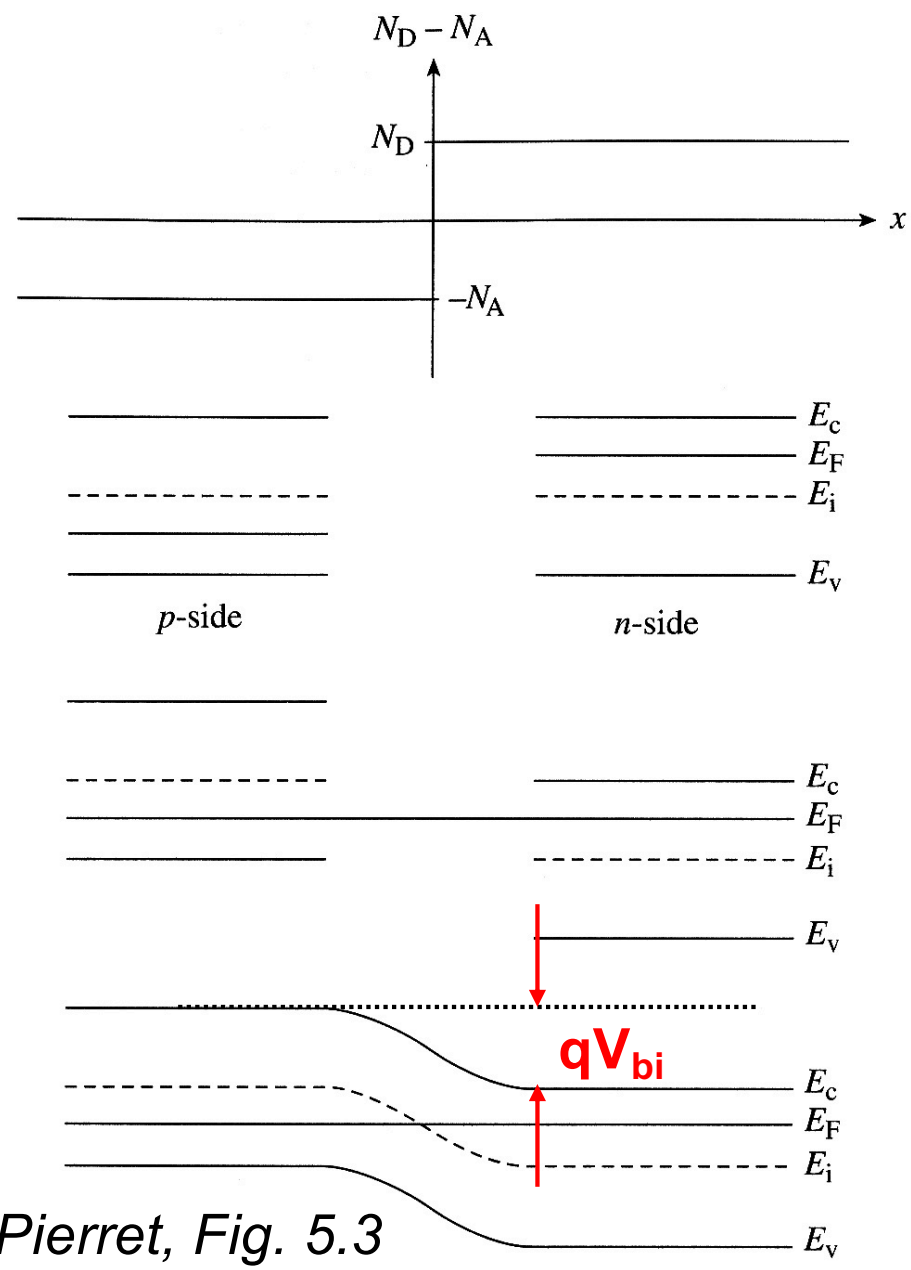
Pierret, Fig. 5.2

Approximations for doping profile in pn junctions:

- **Step Junction**: approximation for ion implantation or **shallow diffusion** into lightly doped wafer
- **Linearly Graded Junction**: approximation for **deep diffusion** in moderately to heavy doped wafer

PN Junction Band Diagram

- **Equilibrium** requires the **Fermi level** to be **constant** across the device
- Regions far away from the metallurgical junction will be unaffected
- Electrons **diffuse** from n- to p-side, holes from p- to n-side, leaving behind unbalanced dopant site charges
- This **space charge** creates an **electric field** (band bending), resulting in a carrier **drift** balancing the carrier **diffusion** (**J_{tot} = 0**)



Pierret, Fig. 5.3

PN Junction Band Diagram

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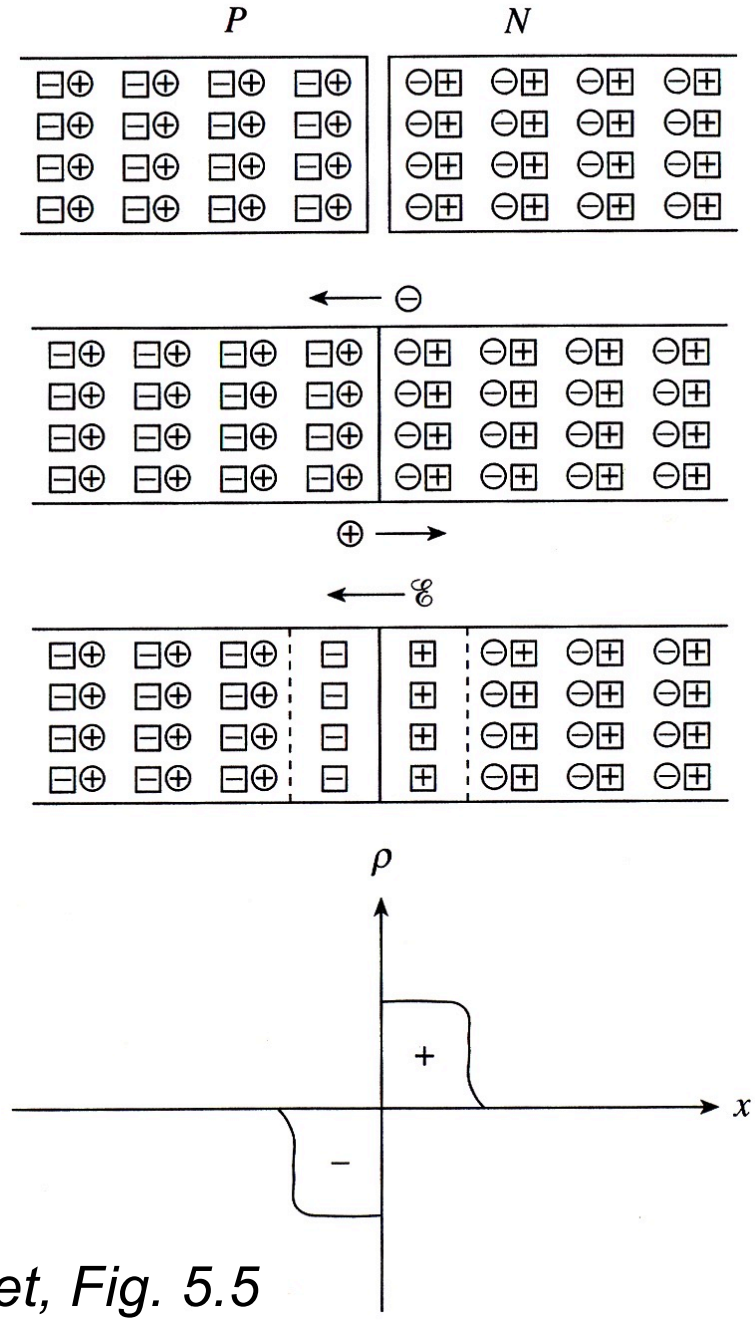
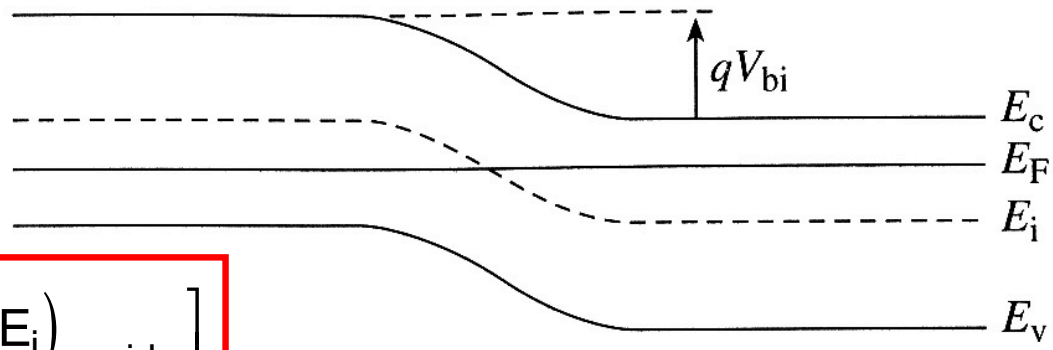


Pierret, Fig. 5.5

PN Junction – Built-in Potential

The **built-in potential** can be calculated from the band diagram:



Pierret, Fig. 5.4

$$V_{bi} = \frac{1}{q}\left[\left(E_i - E_F\right)_{p-side} + \left(E_F - E_i\right)_{n-side}\right]$$

With the results from Chapter 2.5

$$p = N_A = n_i\, e^{(E_i - E_F)/kT} \qquad p-\text{side with } N_A >> n_i$$

$$n = N_D = n_i\, e^{(E_F - E_i)/kT} \qquad n-\text{side with } N_D >> n_i$$

we obtain

$$V_{bi} = \frac{1}{q}\left[kT\,\ln\left(\frac{N_A}{n_i}\right) + kT\,\ln\left(\frac{N_D}{n_i}\right)\right] = \frac{kT}{q}\,\ln\left[\frac{N_A N_D}{n_i^2}\right]$$

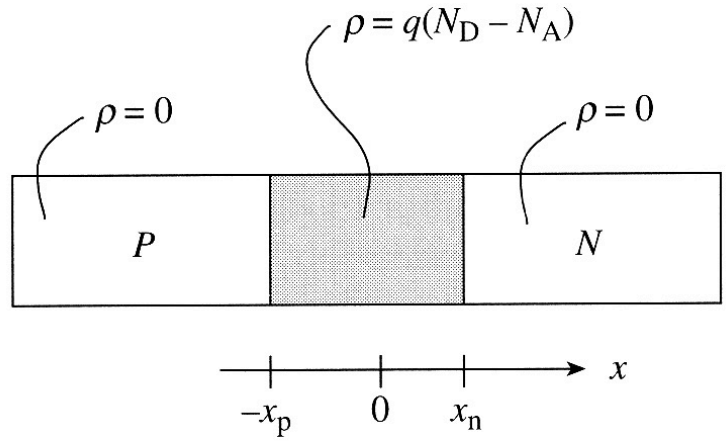
PN Junction – Depletion Approximation

- To obtain quantitative solutions for the electrostatic variables, the **Poisson equation** needs to be solved:

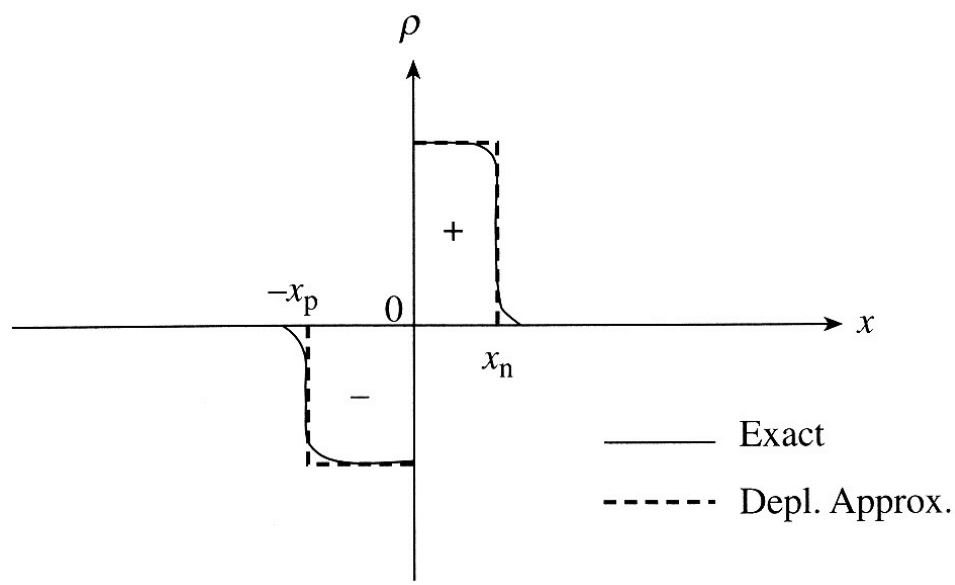
$$\nabla \cdot \varepsilon = \frac{\rho}{K_s \varepsilon_0} = \frac{q\left(p - n + N_D - N_A\right)}{K_s \varepsilon_0}$$

This requires the knowledge of the charge distribution, which itself (n and p) is influenced by the potential

- The **depletion approximation** is often used to approximate the charge distribution:
  1. The carrier concentration n and p is negligible in the depletion region, i.e. for $-x_p \le x \le x_n$
  2. The charge density outside the depletion region is zero



Pierret, Fig. 5.6



Electrostatics

- From charge distribution to electric field:

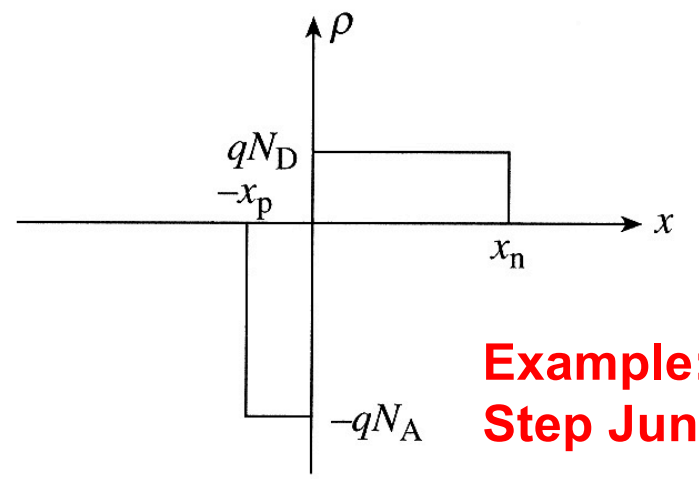
$$\nabla \cdot \varepsilon = \frac{\rho}{K_s \varepsilon_0} \quad \text{or in 1D} \quad \frac{d\varepsilon}{dx} = \frac{\rho}{K_s \varepsilon_0}$$

$$\int_0^{\varepsilon(x)} d\varepsilon = \varepsilon(x) = \frac{1}{K_s \varepsilon_0} \int_{-\infty}^{x} \rho(x)\, dx$$

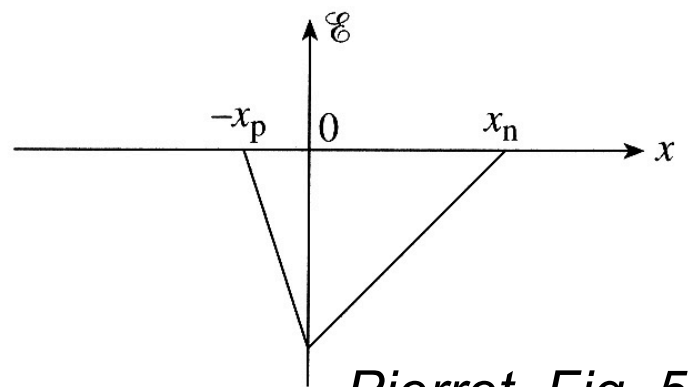
- From electric field to electrostatic potential:

$$\varepsilon = -\nabla V \quad \text{or in 1D} \quad \varepsilon = -\frac{dV}{dx}$$

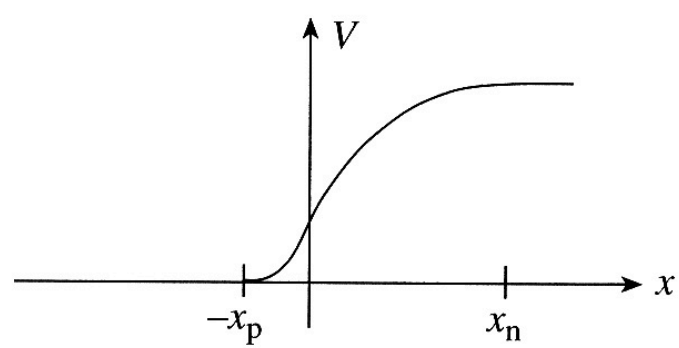
$$\int_0^{V(x)} dV = V(x) = -\int_{-\infty}^{x} \varepsilon(x)\, dx$$



Example: Step Junction



Pierret, Fig. 5.9



3.1.2 Step Junction: $V_A = 0$

- Assuming **depletion approximation** and a **step junction**, the charge density in the depletion region is

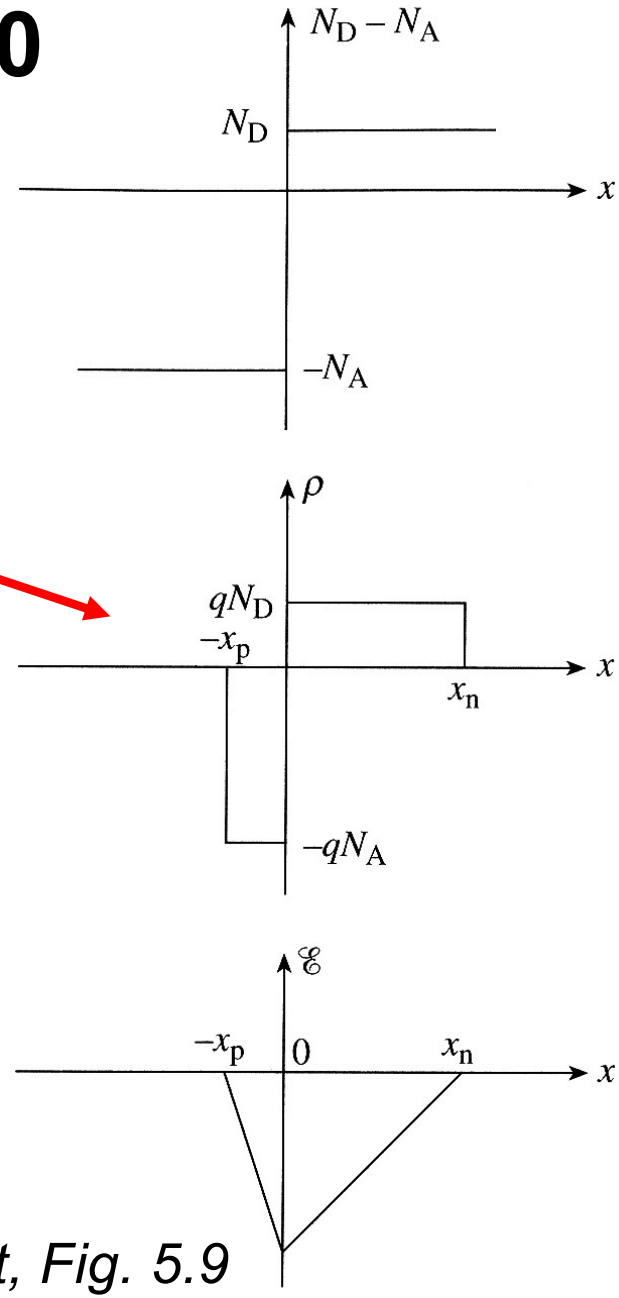
$$\rho = \begin{cases} -qN_A & -x_p \le x \le 0 \\ +qN_D & 0 \le x \le x_n \\ 0 & x < x_p \text{ or } x > x_n \end{cases}$$

- The **electric field** is obtained by integration

$$\int_0^{\varepsilon(x)} d\varepsilon = \frac{1}{K_s \varepsilon_0} \int_{-\infty}^{x} \rho(x)\, dx$$

with the boundary conditions

$\varepsilon(-x_p) = \varepsilon(x_n) = 0$

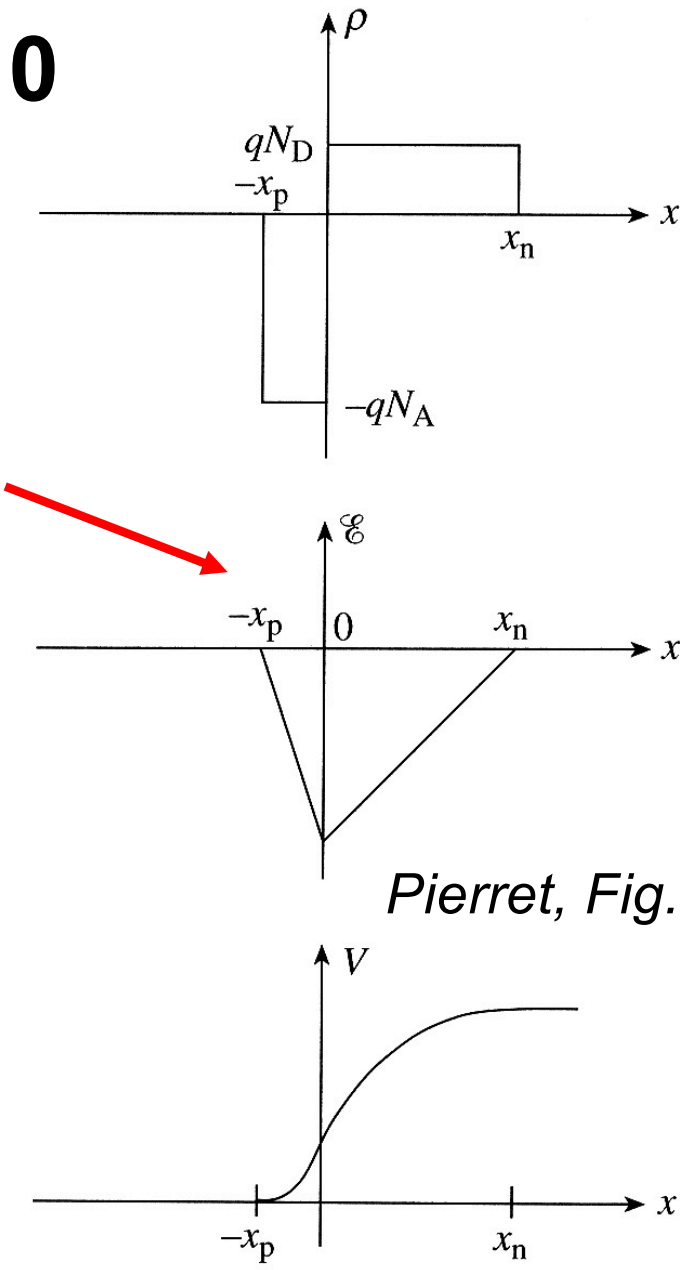


Pierret, Fig. 5.9

Step Junction with $V_A = 0$

- Resulting **electric field**:

$$\varepsilon(x) = \begin{cases} -\dfrac{qN_A}{K_s\varepsilon_0}\left(x_p + x\right) & -x_p \le x \le 0 \\ -\dfrac{qN_D}{K_s\varepsilon_0}\left(x_n - x\right) & 0 \le x \le x_n \end{cases}$$



Pierret, Fig. 5.9

- The **electric potential** is obtained by a second integration

$$\int_0^{V(x)} dV = -\int_{-\infty}^{x} \varepsilon(x)\,dx$$

with the boundary conditions
$V(-x_p) = 0$ and $V(x_n) = V_{bi}$

Step Junction: Charge Considerations

- The electric field must be continuous at x = 0, yielding:

$$-\frac{qN_A}{K_s\varepsilon_0}x_p = -\frac{qN_D}{K_s\varepsilon_0}x_n$$

$$\boxed{qN_A x_p = qN_D x_n}$$

- This constitutes that the **total charge within the depletion region must sum to zero**, i.e. the positive charge on the n-side is balanced by the negative charge on the p-side of the junction
- This overall charge neutrality is a direct consequence of the assumption that the electric field vanishes outside the depletion region, as follows from **Gauss' law**

Step Junction with $V_A = 0$

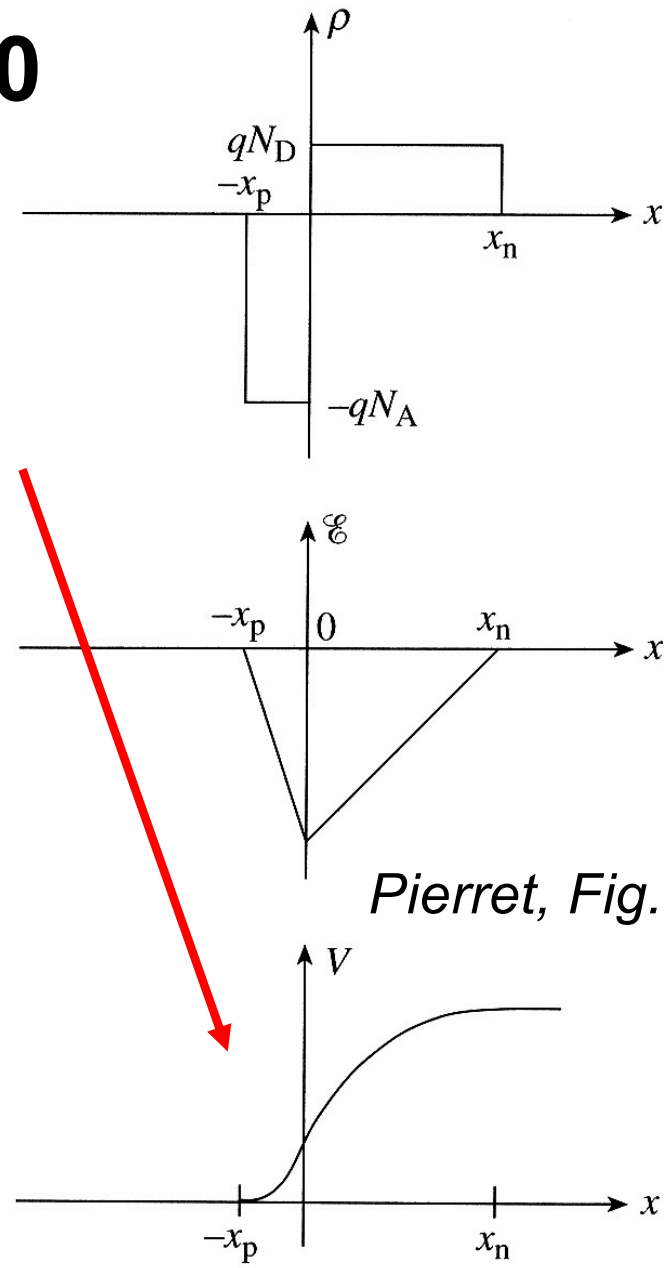
- Resulting **electric potential**:

$$V(x) = \begin{cases} \dfrac{qN_A}{2K_s\varepsilon_0}\left(x_p + x\right)^2 & -x_p \le x \le 0 \\ V_{bi} - \dfrac{qN_D}{2K_s\varepsilon_0}\left(x_n - x\right)^2 & 0 \le x \le x_n \end{cases}$$

- The electric potential is continuous at x = 0, yielding:

$$\frac{qN_A}{2K_s\varepsilon_0}x_p^2 = V_{bi} - \frac{qN_D}{2K_s\varepsilon_0}x_n^2$$

Together with the charge neutrality relationship $N_A x_p = N_D x_n$, this equation can be used to determine x_n and x_p



Pierret, Fig. 5.9

Step Junction – Depletion Layer Width

- We obtain for x_n and x_p:

$$x_n = \sqrt{\frac{2K_s\varepsilon_0}{q}\frac{N_A}{N_D\left(N_A + N_D\right)}V_{bi}}$$

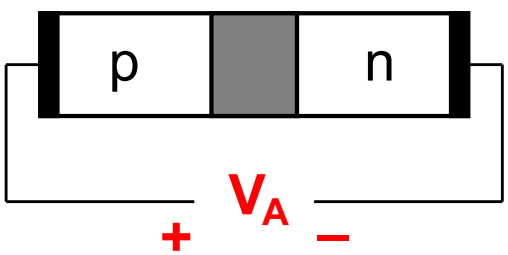
$$x_p = \sqrt{\frac{2K_s\varepsilon_0}{q}\frac{N_D}{N_A\left(N_A + N_D\right)}V_{bi}}$$

- Thus, the depletion layer width W is

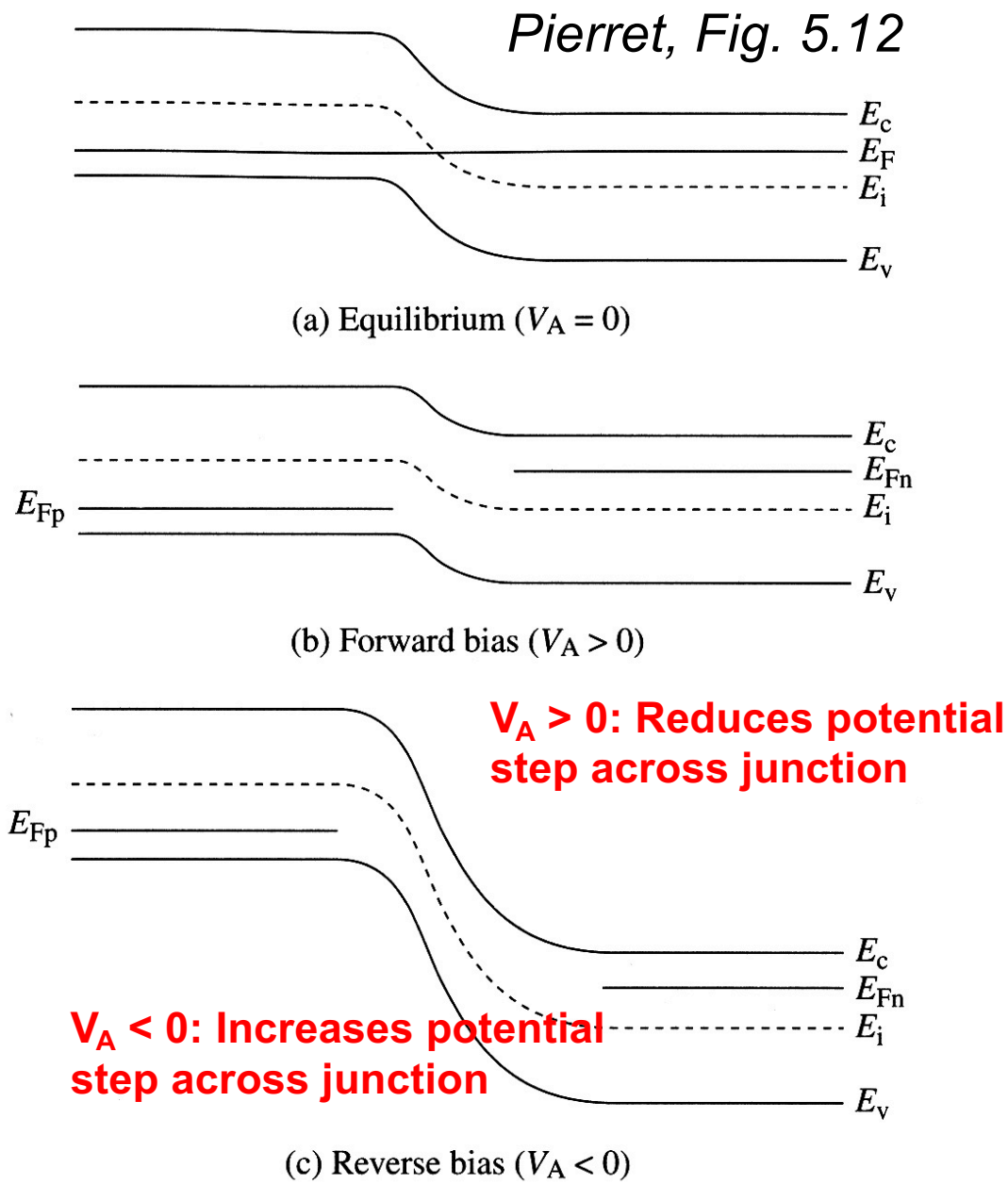
$$\boxed{W = x_n + x_p = \sqrt{\frac{2K_s\varepsilon_0}{q}\frac{N_A + N_D}{N_A N_D}V_{bi}}}$$

Step Junction with $V_A \neq 0$

- Assumption: Externally applied voltage V_A is completely dropped across the depletion region
- I.e., negligible voltage drop at contacts and across the quasi-neutral regions

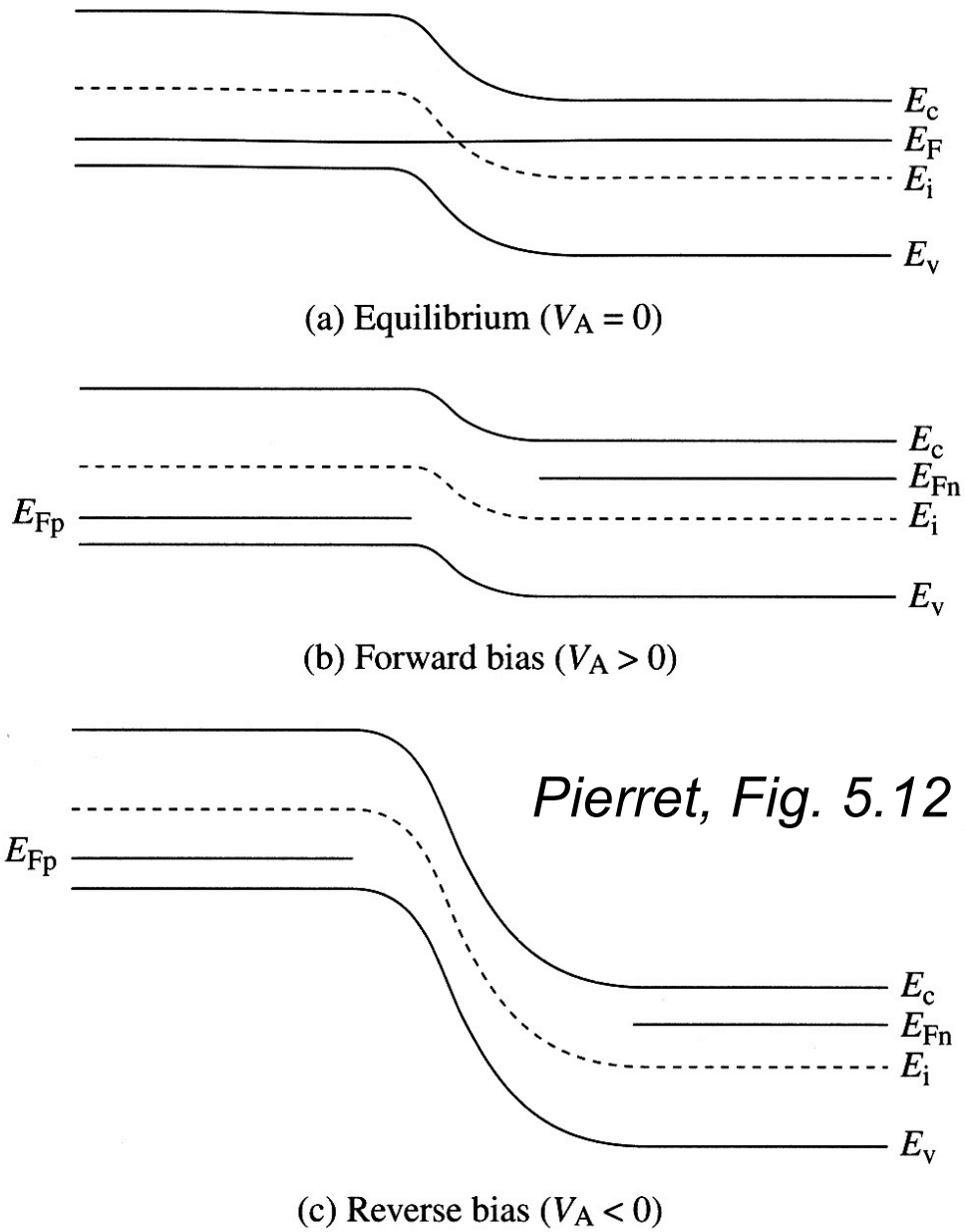


Pierret, Fig. 5.12



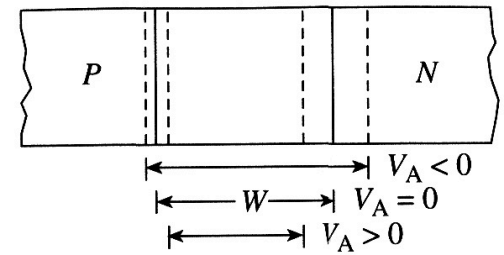
Step Junction with $V_A \neq 0$

- **Forward Bias $V_A > 0$:** Potential drop across the depletion region decreases; **majority carrier diffusion** dominates, resulting in a large forward current flow
- **Reverse Bias $V_A < 0$:** Potential drop across the depletion region increases; **minority carrier drift** dominates, resulting in a small reverse current flow



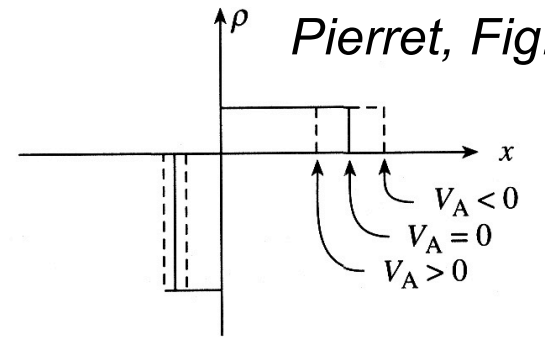
Pierret, Fig. 5.12

Step Junction with $V_A \neq 0$



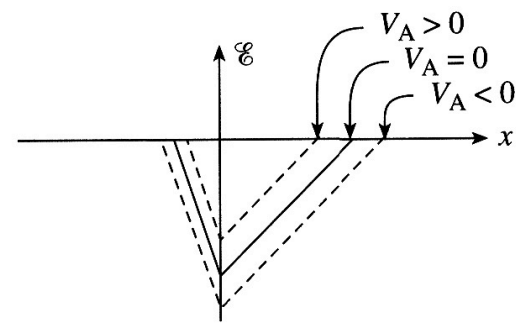
- Bias V_A can be implemented in equations by **replacing V_{bi} with $V_{bi} - V_A$**:

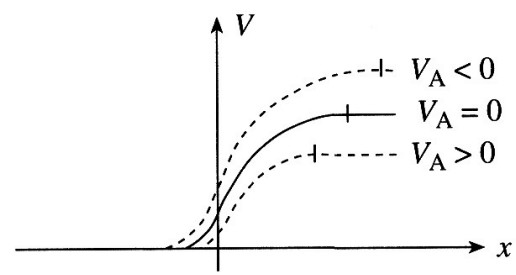
$$W = \sqrt{\frac{2K_s\varepsilon_0}{q}\frac{N_A + N_D}{N_A N_D}\left(V_{bi} - V_A\right)}$$



Pierret, Fig. 5.11

- **Forward Bias $V_A > 0$**
  - Reduced depletion layer width
  - Reduced space charge
  - Reduced electric field
  - Reduced potential drop across junction
- **Reverse Bias $V_A < 0$**
  - Increased depletion layer width
  - Increased space charge
  - Increased electric field
  - Increased potential drop across junction



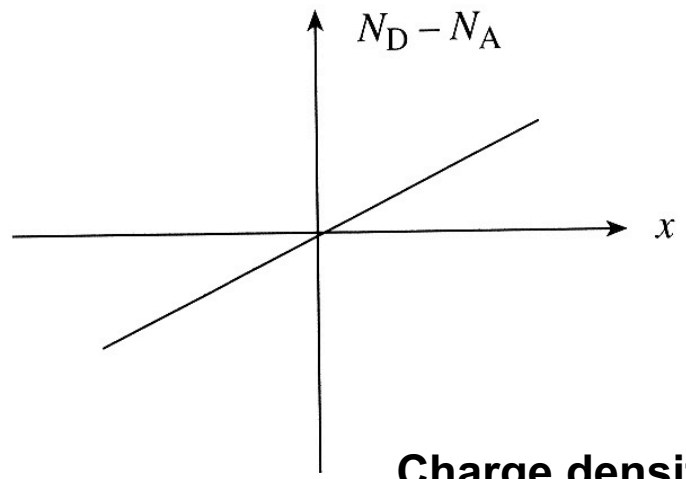


3.1.3 Linearly Graded Junction: $V_A = 0$

- **Doping profile** around metallurgical junction

$$N_D - N_A = ax$$

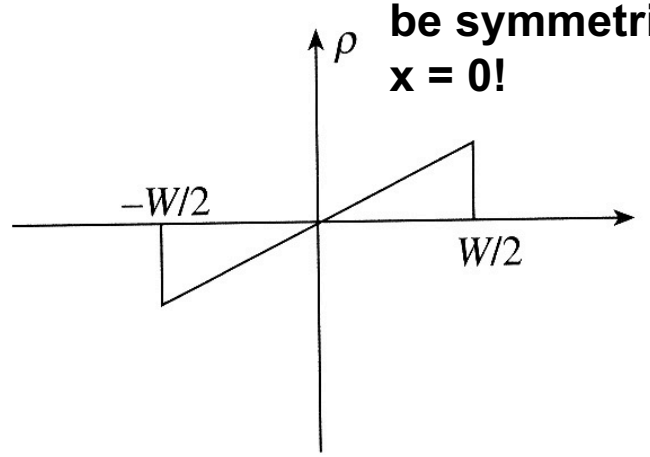
with constant a having units of [cm^{-4}]



- Resulting **charge density**:

$$\rho = \begin{cases} qax & -W/2 \le x \le W/2 \\ 0 & x < -W/2 \text{ or } x > W/2 \end{cases}$$

Charge density must be symmetrical around x = 0!



Pierret, Fig. 5.13

Linearly Graded Junction: $V_A = 0$

- **Electric field** is obtained by integration:

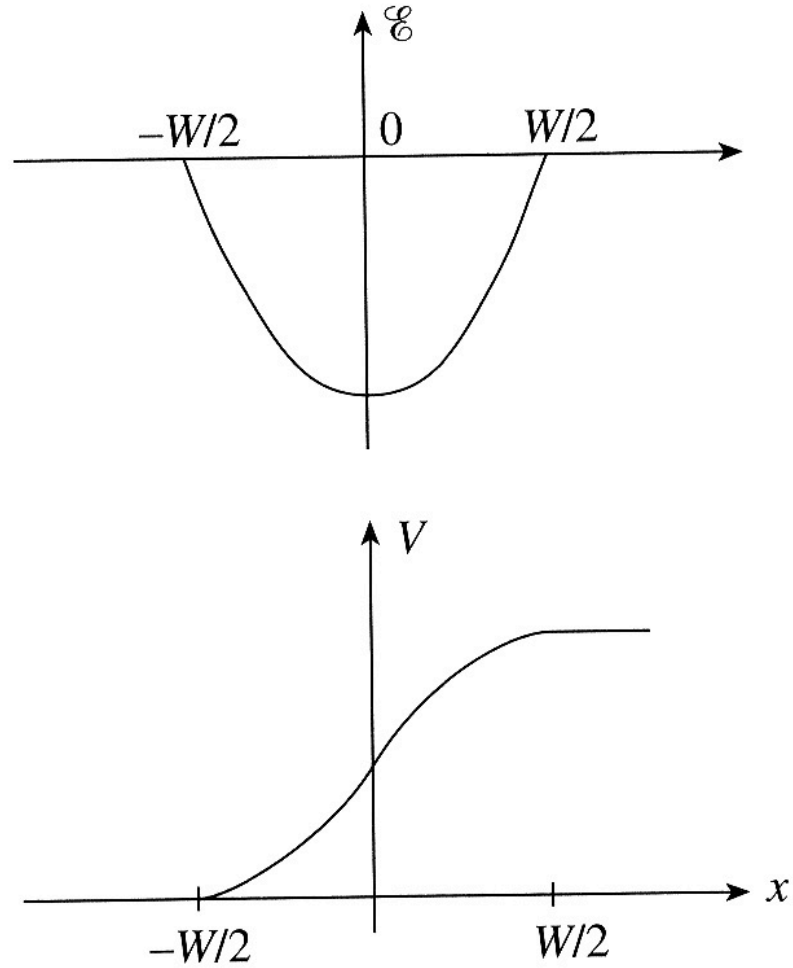
$$\boxed{\varepsilon(x) = \frac{qa}{2K_s\varepsilon_0}\left[x^2 - \left(\frac{W}{2}\right)^2\right]}$$

for $-W/2 \le x \le W/2$

- **Electric potential** is obtained by second integration:

$$\boxed{V(x) = \frac{qa}{6K_s\varepsilon_0}\left[2\left(\frac{W}{2}\right)^3 + 3\left(\frac{W}{2}\right)^2 x - x^3\right]}$$

for $-W/2 \le x \le W/2$



Pierret, Fig. 5.13

Linearly Graded Junction: $V_A \neq 0$

- Depletion layer width of **linearly graded junction** under bias $V_A \neq 0$:

$$W = \left[\frac{12K_s\varepsilon_0}{qa}\left(V_{bi} - V_A\right)\right]^{1/3}$$

- Depletion layer width of **step junction** under bias $V_A \neq 0$:

$$W = \left[\frac{2K_s\varepsilon_0}{q}\frac{N_A + N_D}{N_A N_D}\left(V_{bi} - V_A\right)\right]^{1/2}$$